

Project Implementation Description & Explanation

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Tech Stack

This project relies on two languages, two third party services, and android phones for use. The bulk of the code is written in Kotlin. Kotlin is strikingly similar to Java but more concise and has built-in null safeties but can be run concurrently with Java. Both are supported for development in Android but Kotlin allows for more support from Google including Material 3 which was vital for the aesthetic of this project. The structure used was originally XML but was switched to Compose as it reduced file sizes by allowing functions and UI to be one and the same. Material 3 is a set of guidelines as well as a toolkit/library that allows for clean design that can be dynamic as in this project where it allows the user's phone aesthetic to dictate colors and light mode vs dark mode. Kotlin and Material 3 were used in an Android application to allow the user to interact with this project as an application on their phone. The api requests were handled by a Cloudflare Worker setup and written in Javascript. This was used simply because it was efficient, reliable, scalable, and in testing it was free. The requests were sent to a Twilio telephony service that allowed the call and subsequent text to speech reading of the script to be compliant to FCC regulations about TTS calls. With the user interacting with an Android application using Material 3 aesthetics and Compose structure to send requests via a Cloudflare worker to a Twilio Telephony service the user is able to successfully make a silent emergency report.

UI Design and Flow

On the opening page the user will be welcomed by the title of the application in the top banner (Fig. 1). As they look down they will see a welcome message followed by a prompt to "Select a service below to begin a silent emergency report." Following that is a "How it works" section that explains in three steps the selection, survey, and call flow. Below this at the bottom of the screen are the three choices of emergency responder to choose from based on their emergency. The type of responder was selected to keep the flow simple and quick allowing the user to quickly assess if they need Police, Fire, or EMS. Taping on any of the three options opens up the page and the bar stays persistent in case they made the wrong selection.

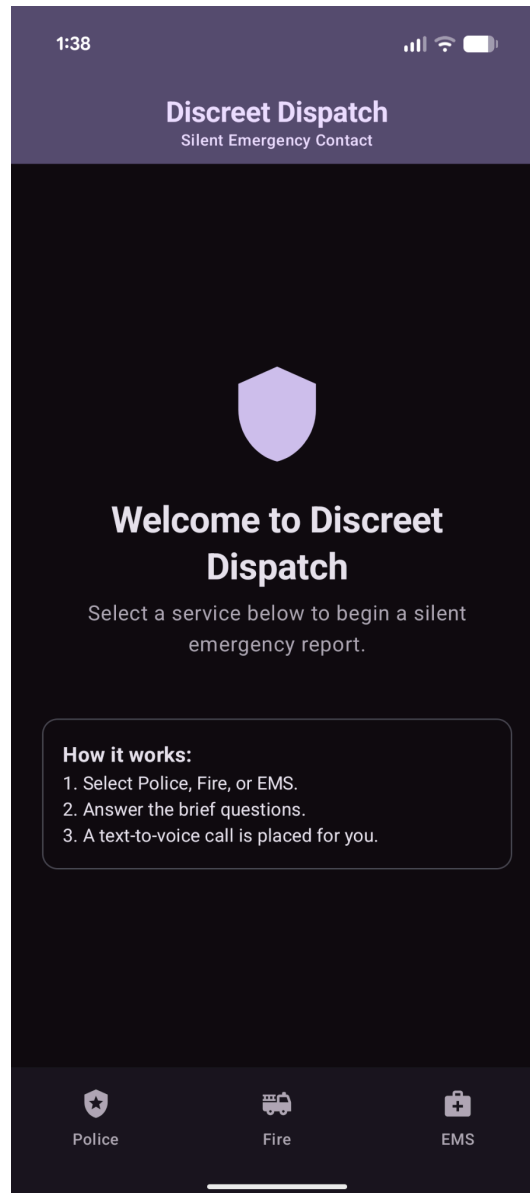


Fig.1 Home Page

Having selected an emergency type/responder the user is given a series of questions similar to those asked by a dispatcher in real life (Fig. 2-4). The set of questions consist of Identifiers both of the victim, perpetrator(if a crime), the situation, safety of the situation/individuals, and finally a section for additional information the user may feel is pertinent. All of these questions are customized to the type of emergency responder/type selected. The questions utilize a series of text fields and radio buttons allowing for either an answer to be filled out or selected based on the type of question. The Textfields have a greyed out prompt that migrates to the top of the text field when filled out and the radio buttons fill in the selection made as solid circles vs the given empty circles. All fields also contain default answers in case they are not or cannot be answered.

1:38

Discreet Dispatch
Silent Emergency Contact

Police Response Request

Enter your name:

Gender:

☐ Male

☐ Female

Enter your address:

Nature of Call:

☐ Theft

☐ Collision

☐ Intruder

☐ Attack

Are you in immediate Danger?

☐ Yes

☐ No

☒ Police

☐ Fire

☐ EMS

Fig.2 Police page Top

1:38

Discreet Dispatch
Silent Emergency Contact

Fire Response Request

Enter your name:

Gender:

☐ Male

☐ Female

Enter your address:

Nature of Call:

☐ Car Fire

☐ Structure Fire

☐ Forest Fire

☐ Hazardous Materials

☐ Other

How intense is the fire?

☐ Low

☐ Police

☒ Fire

☐ EMS

Fig.3 Fire Page Top

1:38

Discreet Dispatch
Silent Emergency Contact

EMS Response Request

Enter your name:

Age Group:

☐ Young: 25 years old or younger

☐ Adult: 25 years to 65 years old

☐ Elderly: over 65 years old

Gender:

☐ Male

☐ Female

Enter your address:

Nature of Call:

☐ Cardiac Issue

☐ Burn (Chemical or Thermal)

☐ Puncture or Laceration

☐ Police

☐ Fire

☒ EMS

Fig.4 EMS Page Top

After Filling out all of the questions the user will see a large “FINISH” button at the bottom of the page (Fig. 5-7). Upon clicking on this button they will be brought to a review page(Fig. 8 & 9). Here, their script that has been formed from the questions they previously answered will be displayed. This allows the user to quickly review their answer before confirming they want to place the call with that script. If there are errors they can simply press back and they will be able to edit their responses or even pick a different emergency responder/type. Once satisfied with their response they will click a button below the script that says “INITIATE EMERGENCY CALL” followed by a warning in red that states tapping the button “will trigger an automated voice call to the dispatch center.” Having clicked the button the user will receive a toast at the bottom of the screen that states: “Call started successfully!” after being taken back to the prompt page(Fig. 10). From here they can either close the app or double check their responses in case an additional call needs to be placed due to new information or more accurate information.

1:39

Discreet Dispatch
Silent Emergency Contact

☐ No

Can you get to safety?

☐ Yes

☐ No

Is anyone else hurt or in danger?

☐ Yes

☐ No

If so, how many persons and what are their conditions?

Describe the offender or offenders:

FINISH

Police Fire EMS

Fig.5 Police Finish

1:39

Discreet Dispatch
Silent Emergency Contact

☐ No

Can you get to safety?

☐ Yes

☐ No

Is anyone else hurt or in danger?

☐ Yes

☐ No

If so, how many persons and what are their conditions?

Give any additional details:

FINISH

Police Fire EMS

Fig.6 Fire Finish

1:39

Discreet Dispatch
Silent Emergency Contact

☐ Yes

☐ No

Can you get to safety?

☐ Yes

☐ No

Is anyone else hurt or in danger?

☐ Yes

☐ No

How many persons and their conditions?

Additional details for EMS:

FINISH

Police Fire EMS

Fig.7 EMS Finish

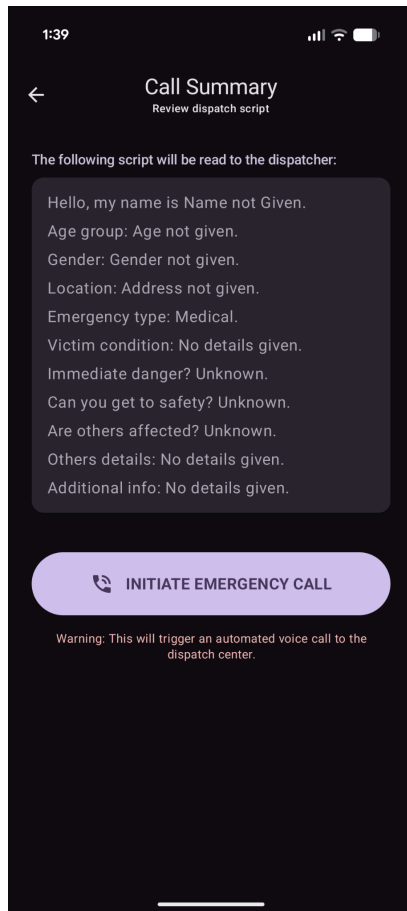


Fig. 8 Blank Script

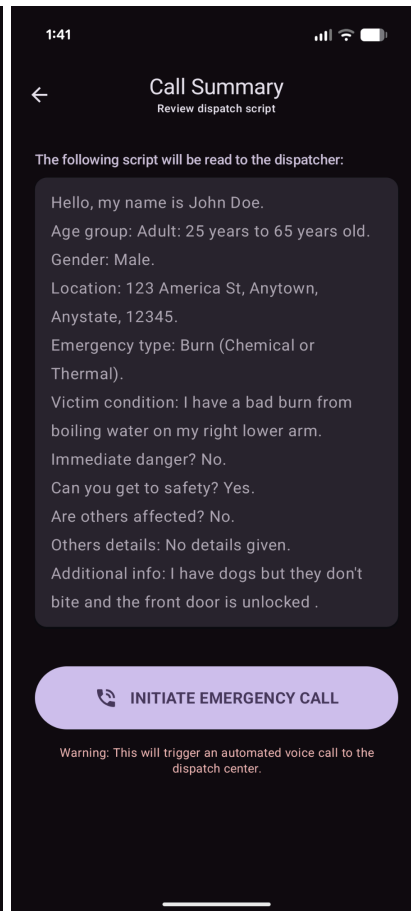


Fig. 9 Filled Out Script

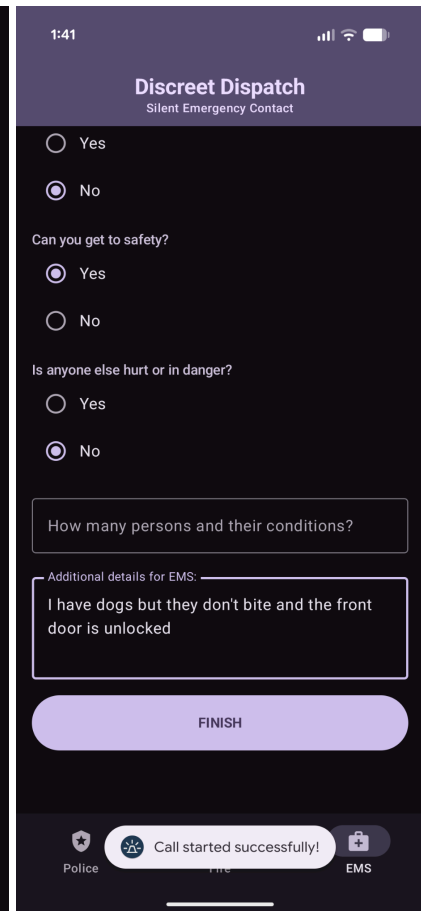


Fig. 10 Completion Toast

Future Implementation and Additions

For the application to become a fully functional product rather than a proof of concept there would need to be the ability to give the call a route, the ability to record calls in more detail, and the ability to report/identify failures remotely. To begin the application as it relies on the user to know the location of the emergency. Further development of the project should make room for the application to utilize GPS services like Google maps or ATAK-Civ to allow for the user to more quickly allow the phone to use GPS to identify location. Doing so could allow for unsure users and users under duress to better identify their location. Beyond identifying the user's location it must also route the call to the respective emergency dispatch service. As the application is merely a proof of concept it only calls one number designated for testing as calls to actual dispatch would be illegal when no emergency is present. Beyond location and routing the current design relies on the third party services for recording information on the calls with Cloudflare recording the requests and Twilio recording when the calls were replaced as well as their duration. Further development should implement a function and secure database that allows the script, request status, call target, call time, and call duration to be recorded for later troubleshooting or if subpoenaed by a court. Lastly the current design handles some errors but does not allow for reporting, reporting in conjunction with the suggested database would allow for ease at identifying issues and resolving them. If these additions are implemented the app

would be ready for real world use as the services used allow for scalability, at a price of course, and could be used at large scale once published.